

NILOY JANA

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EDUCATION

St Thomas College of Engineering and Technology - B.Tech Computer Science and Engineering [2027]
South Point High School - (CBSE) X | (CBSE) XII [physics, chemistry, mathematics, biology] [2021]

EXPERIENCE

AI/ML Engineer Intern | KreupAi Technologies LLC | April 2026 - July 2026 | [Certificate](#)

- Developed a multi-agent LLM system using Python and FastAPI, orchestrating 25 AI agents to automate enterprise QA and Finance workflows, monitoring performance across ~2M tokens and reducing API costs by ~18%.
- Built an AI-powered invoice processing agent using NLP and document parsing to perform automated 3-way matching and fraud detection, reducing manual processing by ~75%.
- Engineered a fault-tolerant LLM Gateway with multi-provider failover and Circuit Breaker pattern achieving 99.2% uptime, integrated with a PII detection and redaction pipeline sanitizing 8 classes of sensitive financial data.
- Implemented a RAG (Retrieval-Augmented Generation) pipeline using text embeddings and pgvector in PostgreSQL for semantic search and context-aware AI responses.

Artificial Intelligence Intern | Pratinik Infotech | March 2026 - May 2026 | [Certificate](#)

- Built a predictive maintenance ML model in Python using sensor data preprocessing, feature engineering, and anomaly detection, achieving ~89% failure prediction accuracy and reducing equipment downtime by ~30%.
- Developed a real-time monitoring and alerting pipeline with an interactive health metrics dashboard, enabling proactive maintenance recommendations across equipment failure and anomaly detection use cases.

PROJECTS

ARIA - Multi-Agent AI Financial Advisory | Python | RAG (FAISS vector store)|REST API | SQLite [GitHub](#)

- Developed ARIA, a full-stack, privacy-first AI financial advisory web application (FastAPI + Vanilla JS) integrating a multi-agent LLM reasoning system (Ollama/Groq) to deliver personalized, tool-driven financial planning across 8 domains — including planning, tax optimization, portfolio analysis, and couples' finance management.
- Engineered a RAG-powered document analysis system using FAISS vector store with 800-token chunking and top-4 semantic retrieval, paired with a dual AI-provider architecture supporting fully offline local inference — eliminating cloud data dependency unlike conventional AI finance tools.

Opsira - AI Eye Disease Detection System | Python | TensorFlow/Keras | OpenCV | Flask [GitHub](#)

- Developed a computer vision and deep learning based eye health screening system using CNN (EfficientNetB0) with 2-phase transfer learning, classifying 4 ocular conditions (Cataract, Conjunctivitis, Jaundice, Healthy) from 224×224 medical images with data augmentation and balanced class weighting.
- Implemented Explainable AI using Grad-CAM heatmaps to visualize model decision regions, and built a rule-based image processing pipeline using OpenCV extracting 5 physiological biomarkers (redness, yellowness, cloudiness, brightness, contrast) with a hybrid boosting layer adjusting prediction confidence by up to ±60% and 3-tier severity classification (LOW/MODERATE/HIGH).

Sign Language detector - Real Time Hand Gesture Recognition System |Python| OpenCV |Scikit-Learn [GitHub](#)

- Built a real-time hand gesture recognition system using MediaPipe to extract 21 hand landmarks (42 normalized features) from a live webcam feed, classified via a linear SVC trained on an 80/20 split achieving ~97% accuracy, supporting 2-hand simultaneous detection with live confidence scoring.

TECHNICAL SKILLS

Languages - Python , Java

Frameworks - TensorFlow, PyTorch, Scikit-learn, Keras, NumPy, MediaPine, OpenCV, LangChain

AI/ML - Generative AI, RAG , Deep Learning , NLP (Transformers, BERT, LLMs)

Web - HTML, CSS, REST APIs , JavaScript , React, FastAPI, Flask, Node.js

Databases - MySQL, SQLite

Tools - Git, GitHub

CERTIFICATIONS

- Infosys Certified in Deep Learning** - Demonstrating proficiency in AI/ML concepts and practical model development [April 2026]
- IBM AI Engineering Professional Certificate** - Comprehensive training in Machine Learning, Deep Learning, Generative AI, LLMs, RAG, Transformers, and AI Agent Development [May 2026]

ACHIEVEMENTS

- AI Hackathon - Indian Institute of Technology Delhi: Ranked Top 8 out of 1400+ participants; built a hybrid LLM + rule-based AI agent for an Among Us simulation with BFS navigation, chat analysis, and strategic voting logic.
- Led the organization and execution of a College-Level Coding Competition, managing participants and event coordination
- Achieved a place in semi finals among 1100 participant in ET-AI HACKATHON hosted by ECONOMIC TIMES